

Weimo

Multimodal BCI for Autonomous Wheelchair Navigation

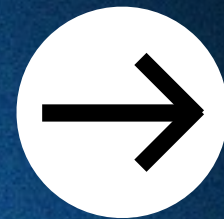


Aaron Pilapil, Andrew Kim, Anthony Saldana, Chengyi Li, Devin Seberino, Diego Tyner, Joshua Wei

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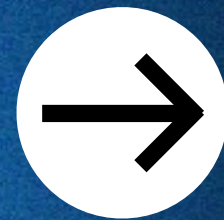
Project Definition

Motivation and problem solved



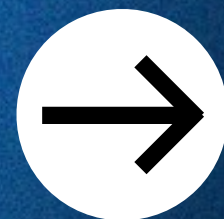
EEG

Paradigm, Processing, Classification



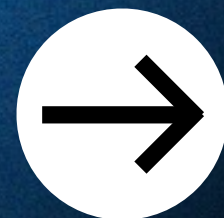
Hardware & Software

Motors, Lidar, VLM, Eye Tracking



Conclusion

Next immediate steps



Project Overview

The Problem:

Project focus: Address mobility limitations for those affected by neurodegenerative disorders

We focus on conditions where users retain eye gaze and cognitive function but lose voluntary motor control, such as late stage ALS or upper spinal damage

Existing solutions demand high focus from the user in order to provide continuous inputs

Furthermore, movement is segmented and precise adjustments are difficult

We propose a multimodal BCI for users to intuitively navigate their environment



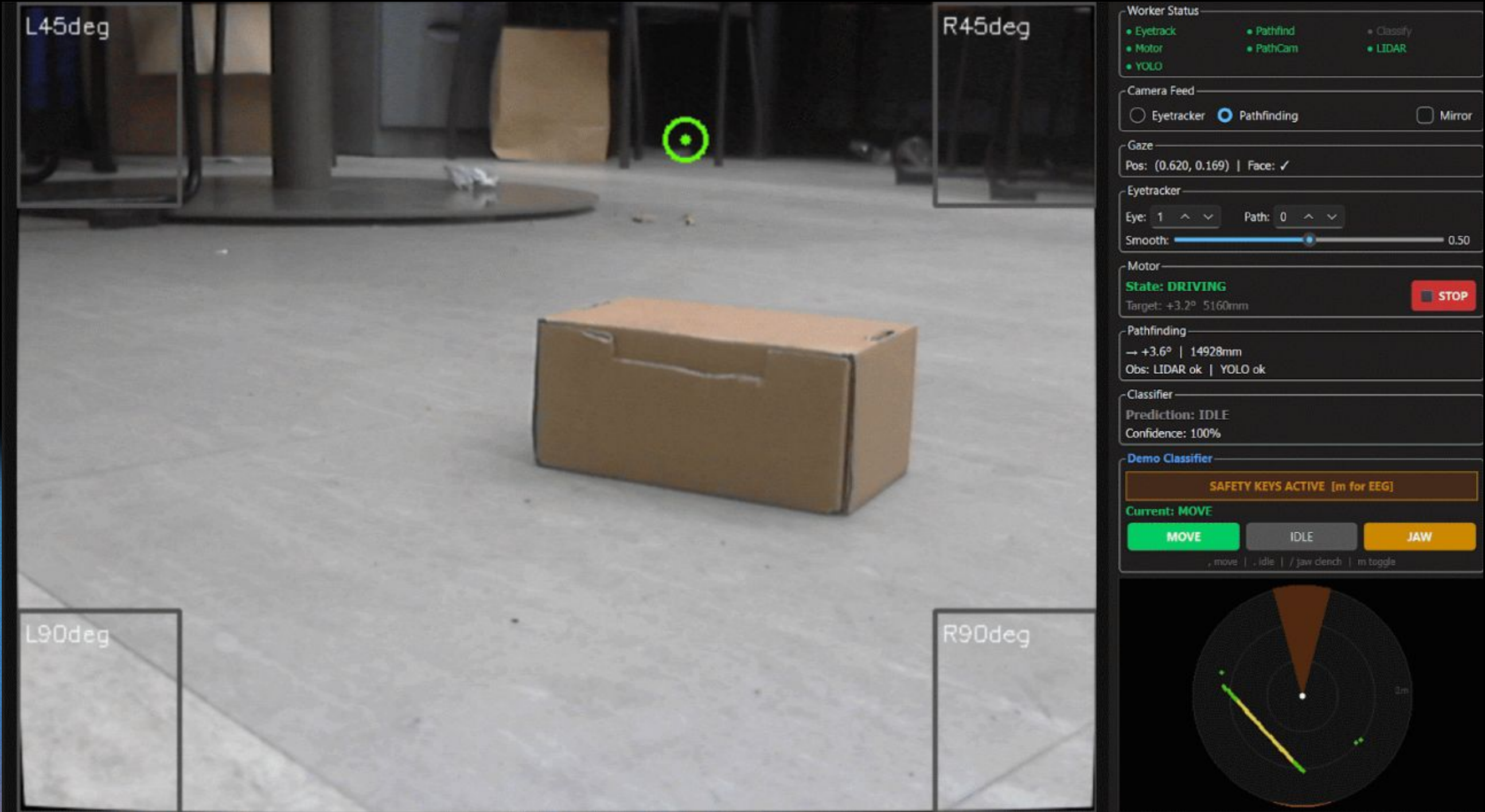
Multimodal BCI Wheelchair Prototype

Reduces cognitive load
utilizing lidar data, face
tracking, and EEG motor
imagery

Our Solution



Example UI



Architecture and EEG

Thank You to our Sponsor

WEARABLE 
Sensing



Architecture

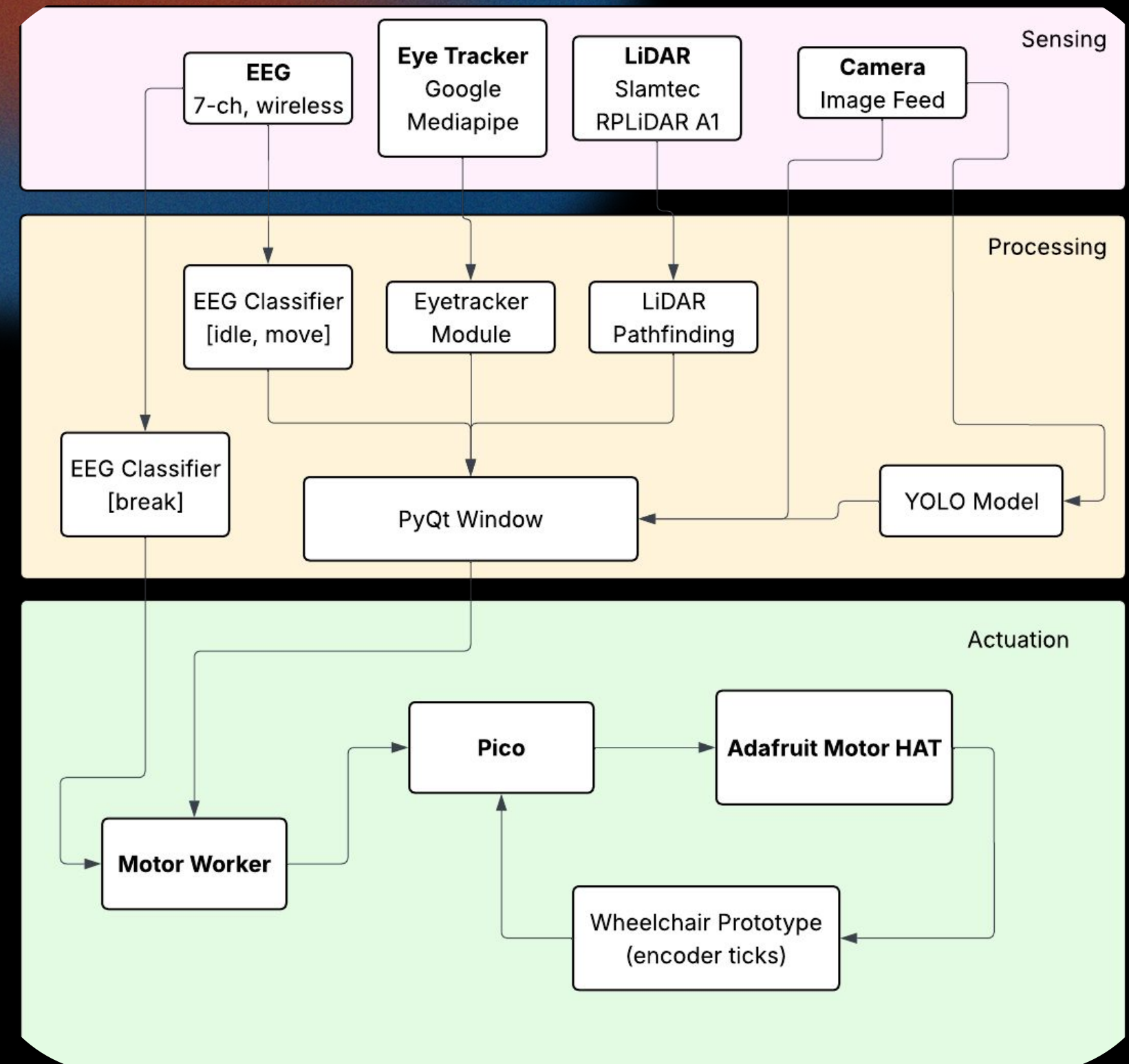
Utilize inputs from multiple modalities:
face tracking, lidar, and EEG

Decoding:

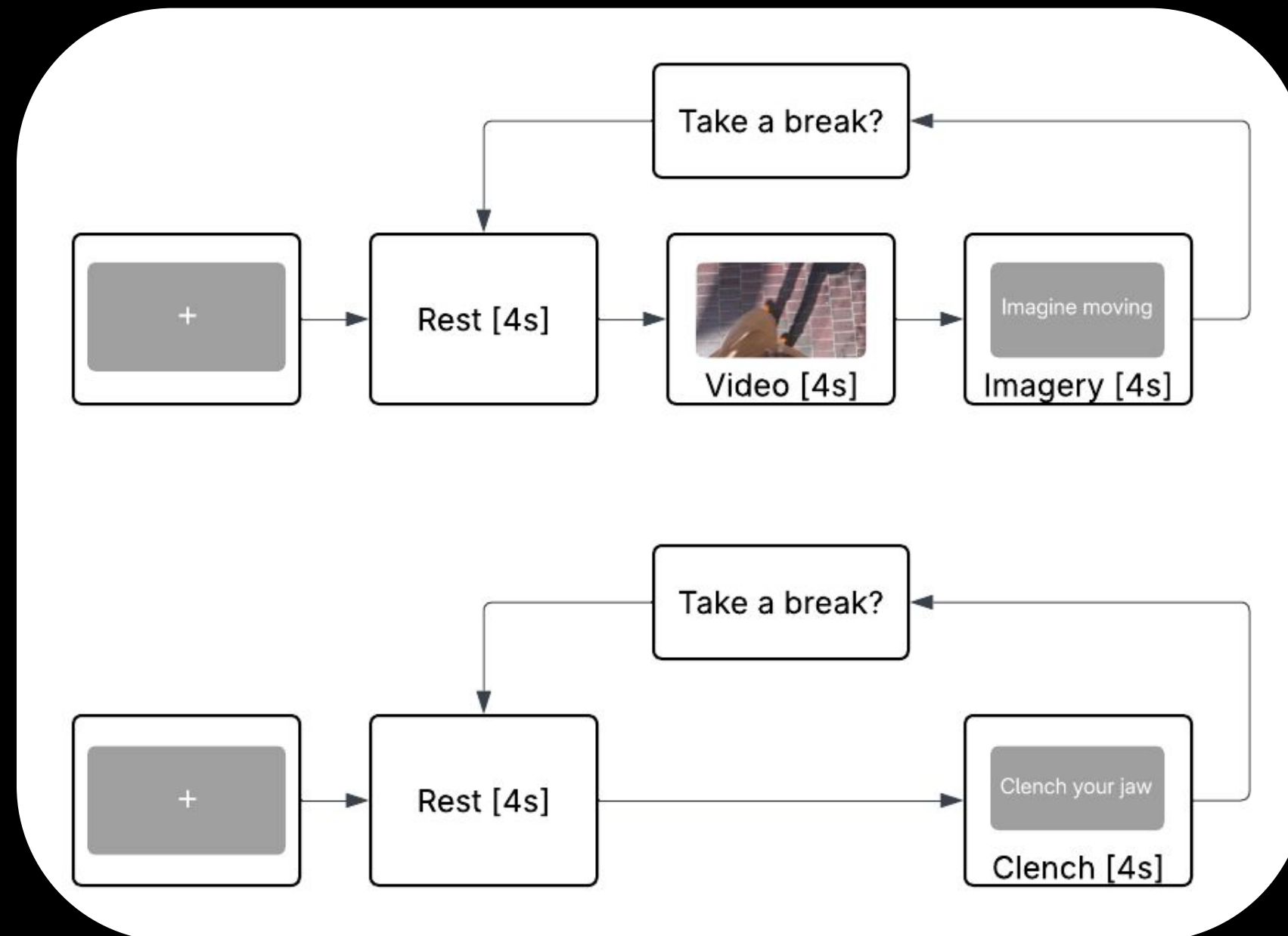
Lidar → Environmental Mapping
Face tracking → User selects a
location

EEG → Decode motor imagery
YOLO v8 → Obstacle detection

Laptop acting as the primary
processor



Training Paradigm



- A few practice trials
 - Motor imagery: 36 trials
 - Jaw clench: 36 trials
- Idle: No motor imagery or jaw clench
- User can take breaks
- Ran several training sessions to get hundreds of data points
- Electrode placements at FZ, F4, C4, P4, P3, C3, F3, P2

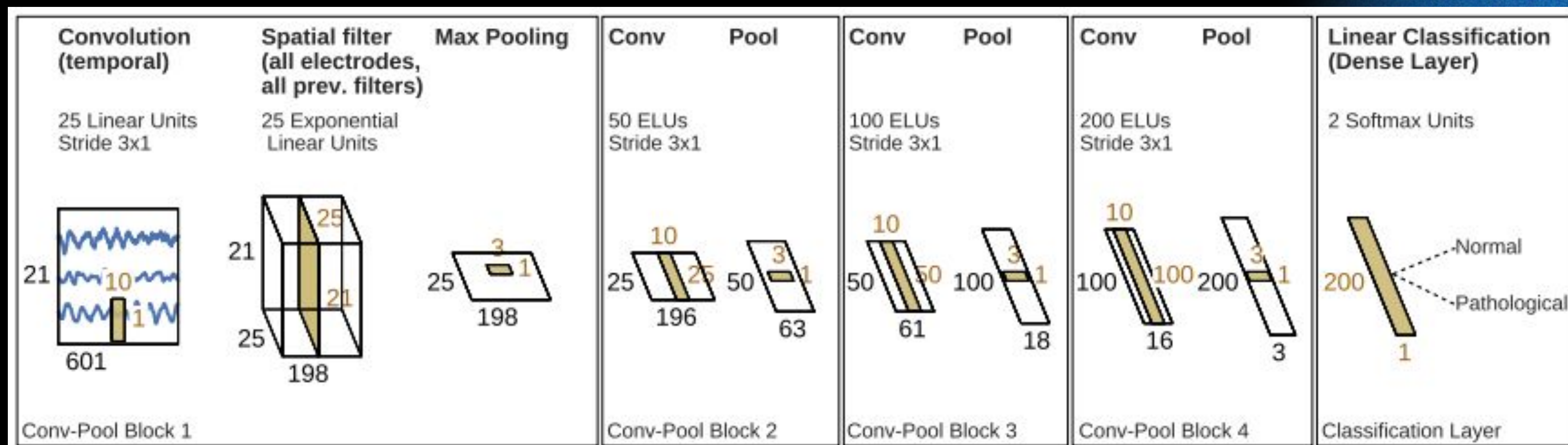
EEG/EMG Processing

- We apply the following EEG processing
 - Notch at 60Hz
 - Beta bandpass
 - Reference to the average
- SVM (Jaw Clench)
 - Log-mean bandpower, 5 bands $\times$ 8 ch = 40 features
- DeepConvNet (Motor Imagery)
 - Z-score normalization

Classifier Architecture

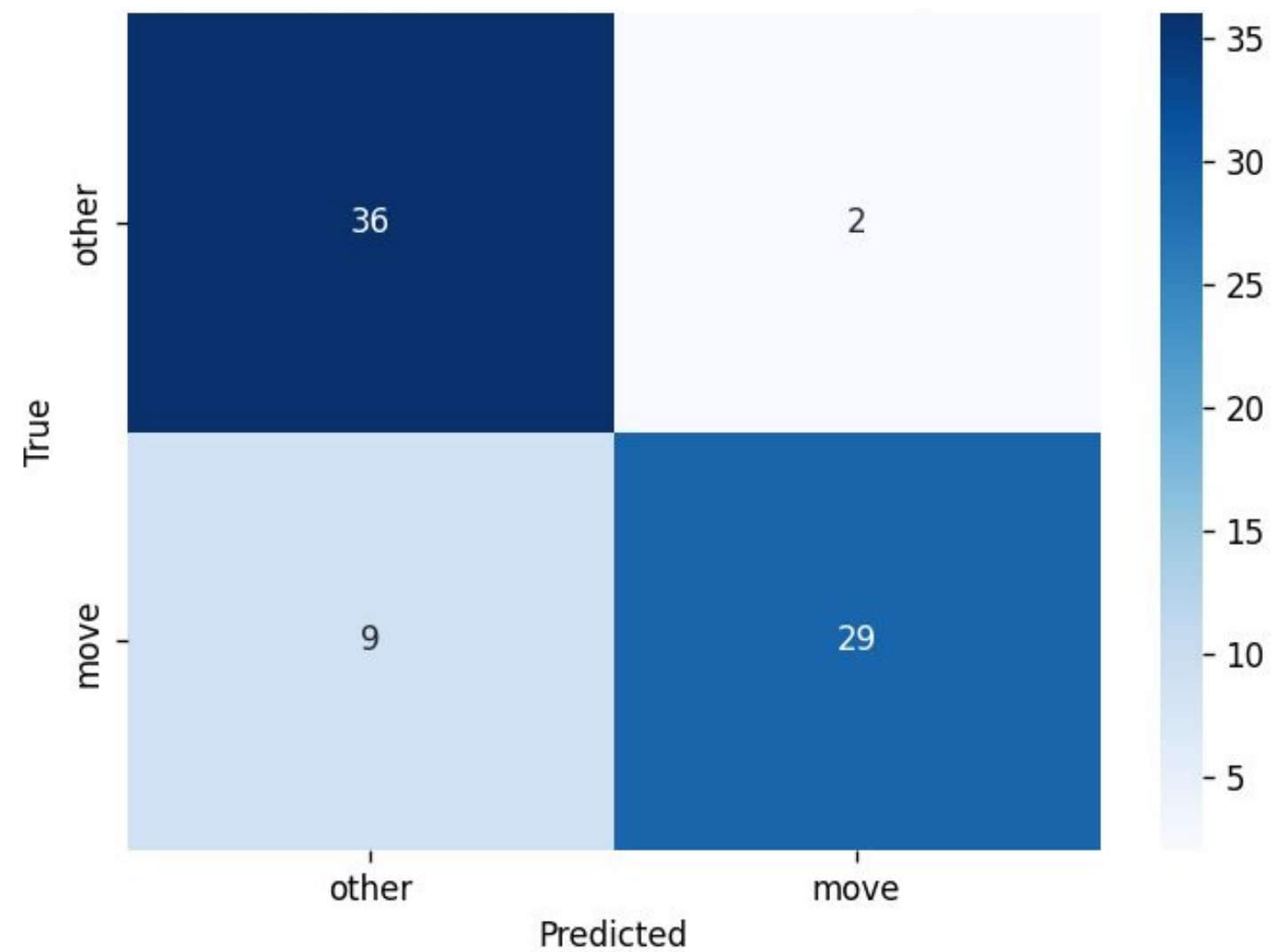
Architecture Inspired by "Deep learning with convolutional neural networks for decoding and visualization of EEG pathology" by Schirrmeister et al 2017.

- ~270k params

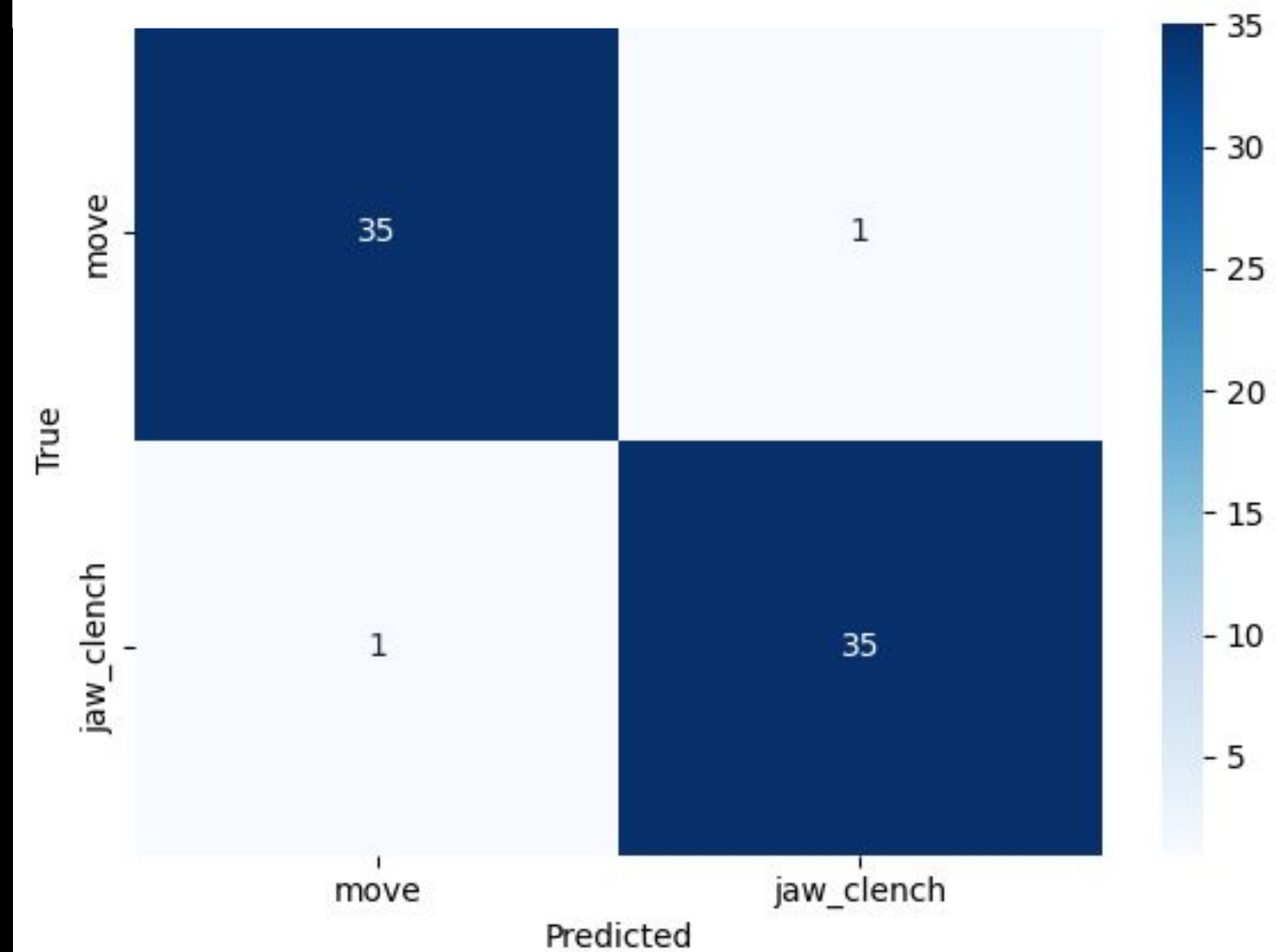


Classifier Results

DeepConvNet (85.54% accuracy)



SVM (97.41% accuracy)



Hardware and Software

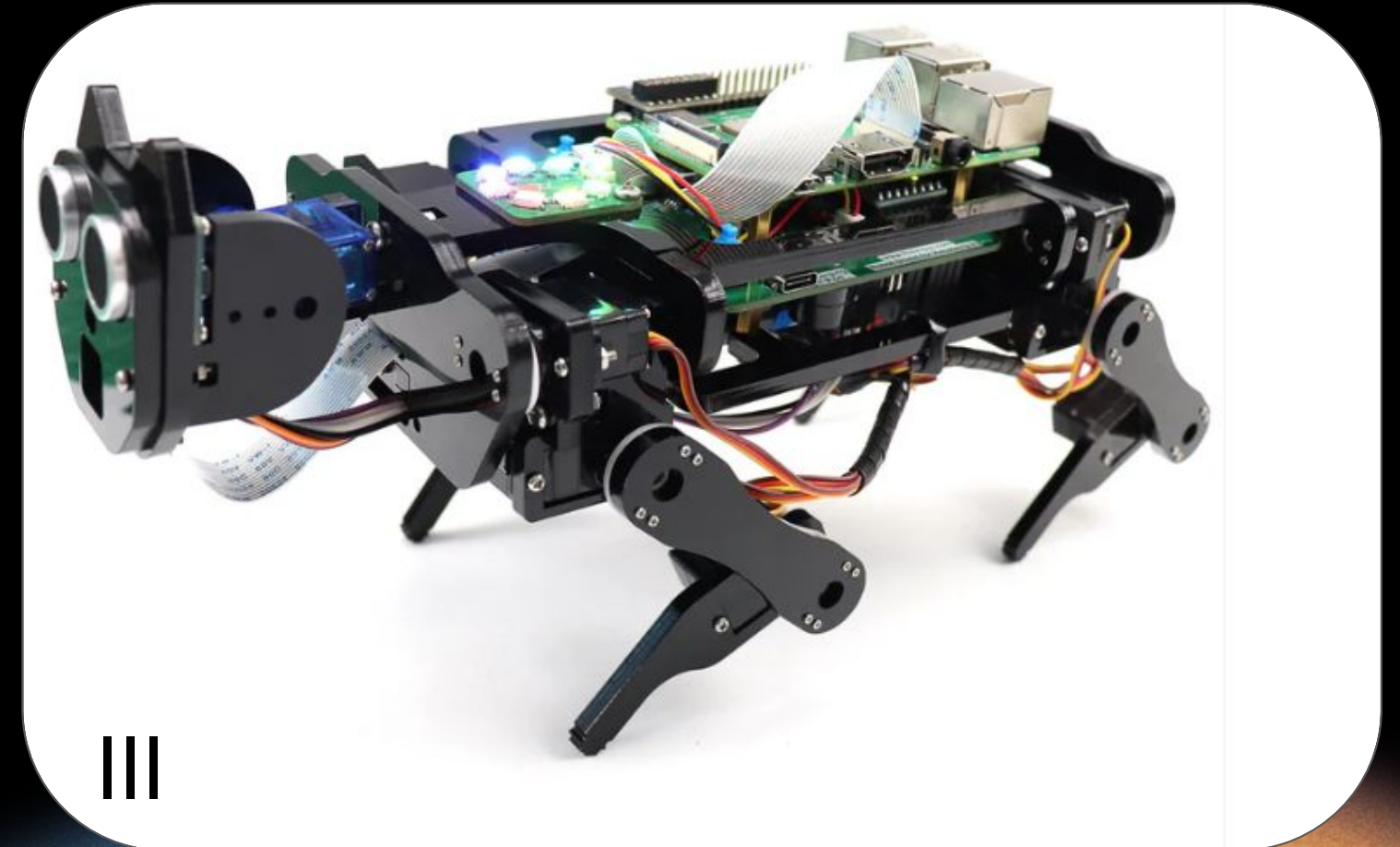
Platform Selection

Evaluated Options:

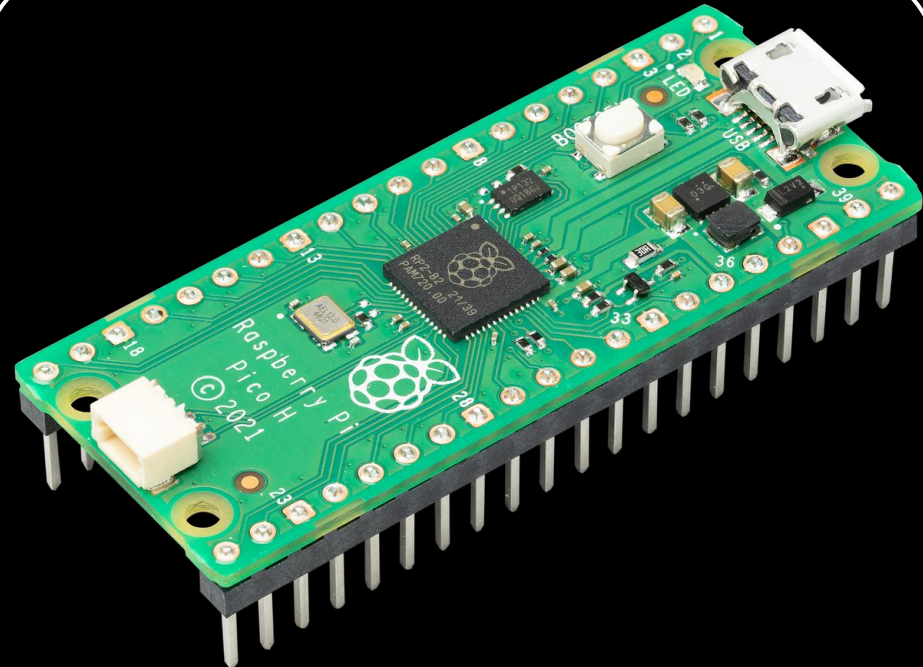
- I. Electric wheelchair
 - Out of budget, including second hand options
- II. Motorizing a manual wheelchair
 - Possible, but beyond the scope of this project
- III. Non car based systems
 - Inconsistent with motor drivers
 - Difficult to add modalities

Selected Platform:

- IV. Omniwheel RC Car
 - Compatible with other hardware ☐
 - Within budget ☐

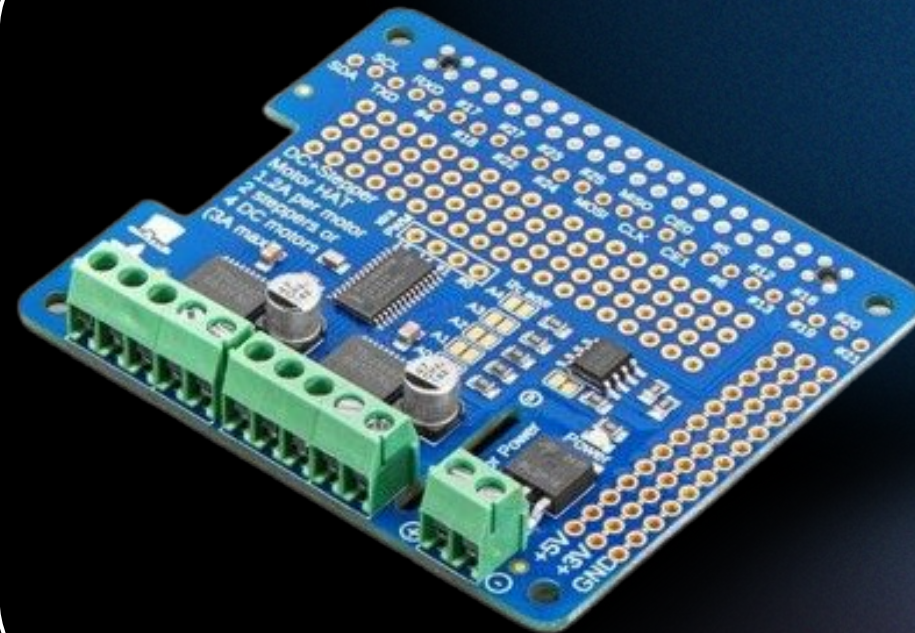


Motor Architecture



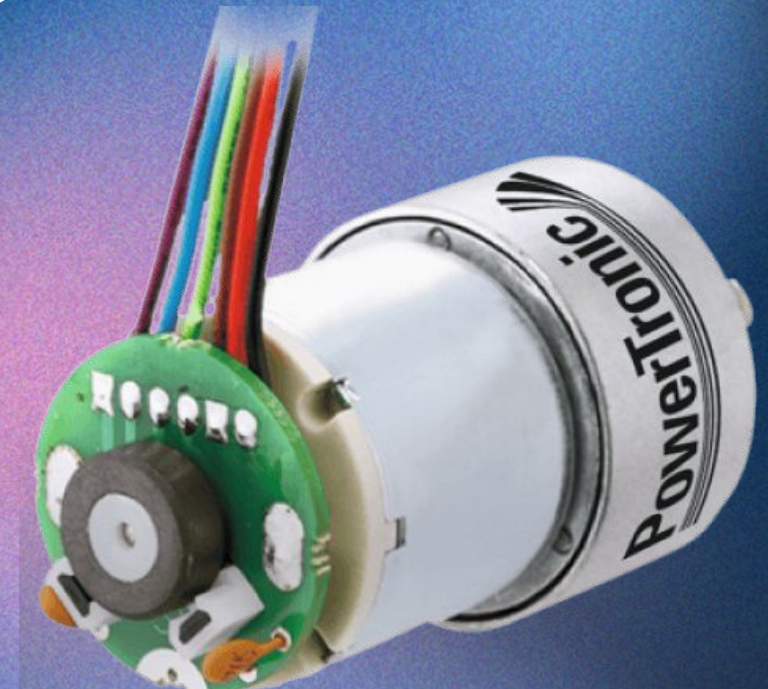
Raspberry Pi Pico

Core Processor for motors. Receive real time data from motors to apply a PID controller that reliably and smoothly guides the car to its destination



Adafruit Motor HAT

Motor Driver that precisely controls the power we draw from the 7.4V Li-Ion battery. Contains the H-bridge logic that allows for full rotational control of the motors, forwards and reverse.



Encoder Motors

Motors that feed the pico ticks, which translate how far the wheels have spun to binary bits that the microcontroller can read

Front Facing Camera

- Logitech C270 camera mounted at the front of the car
- Displays environment to user
- Responsible for obstacle detection



Lidar

Chosen Hardware: Slamtec RPLiDAR A1

- Cost efficient ✓
- Sufficient documentation ✓

Performance Goals:

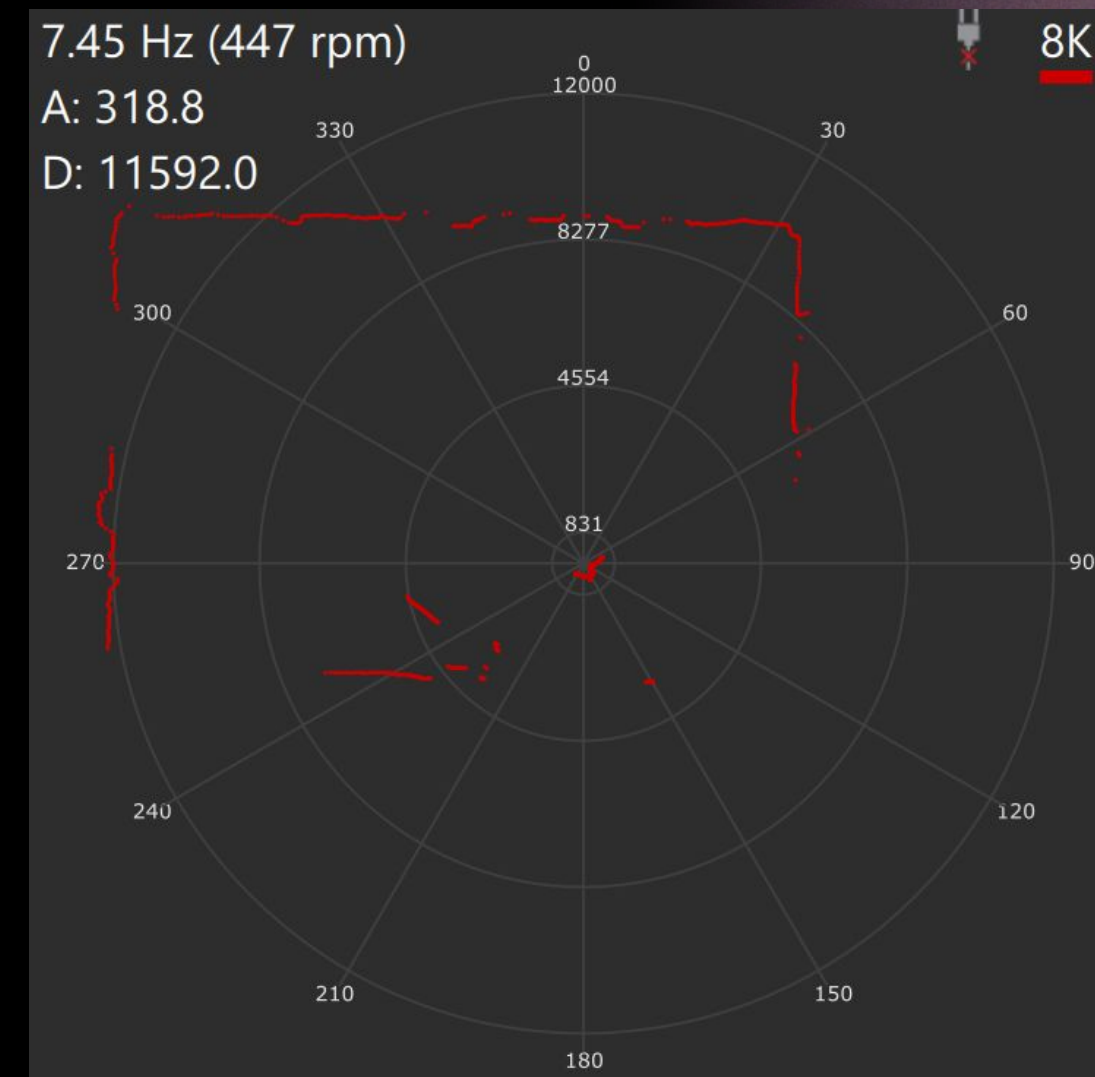
- Provide user with sufficient environment awareness
- Identify objects not pictured in camera's FOV



Lidar

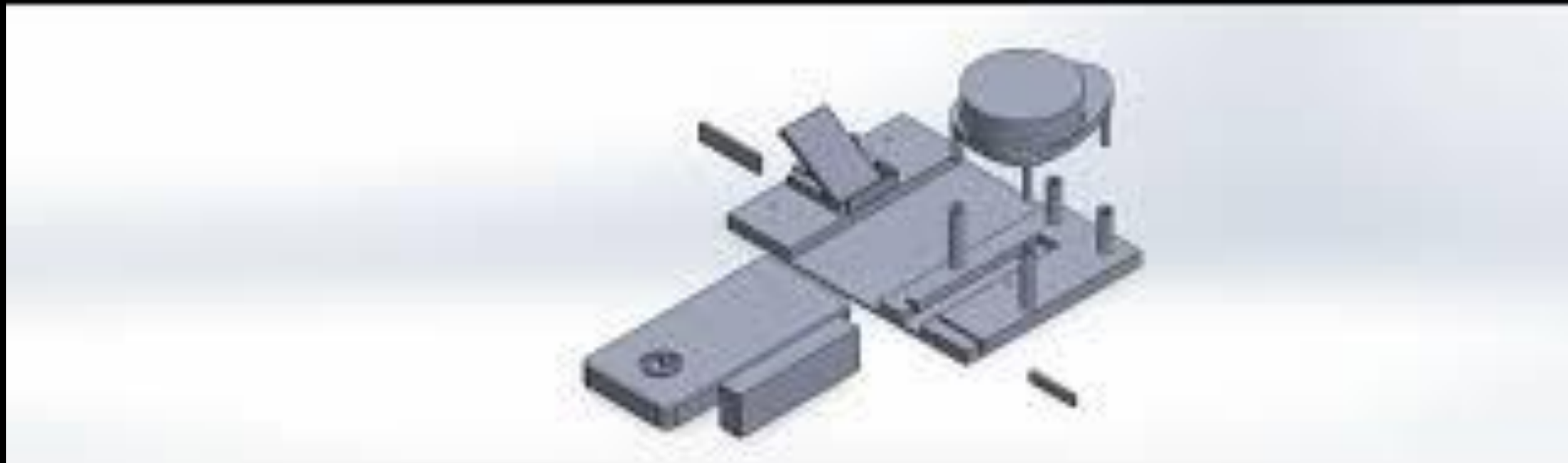
Visualization Goal:

- Collect scans
- Apply convolution to smooth nearby points
- Present 2D occupancy map to user



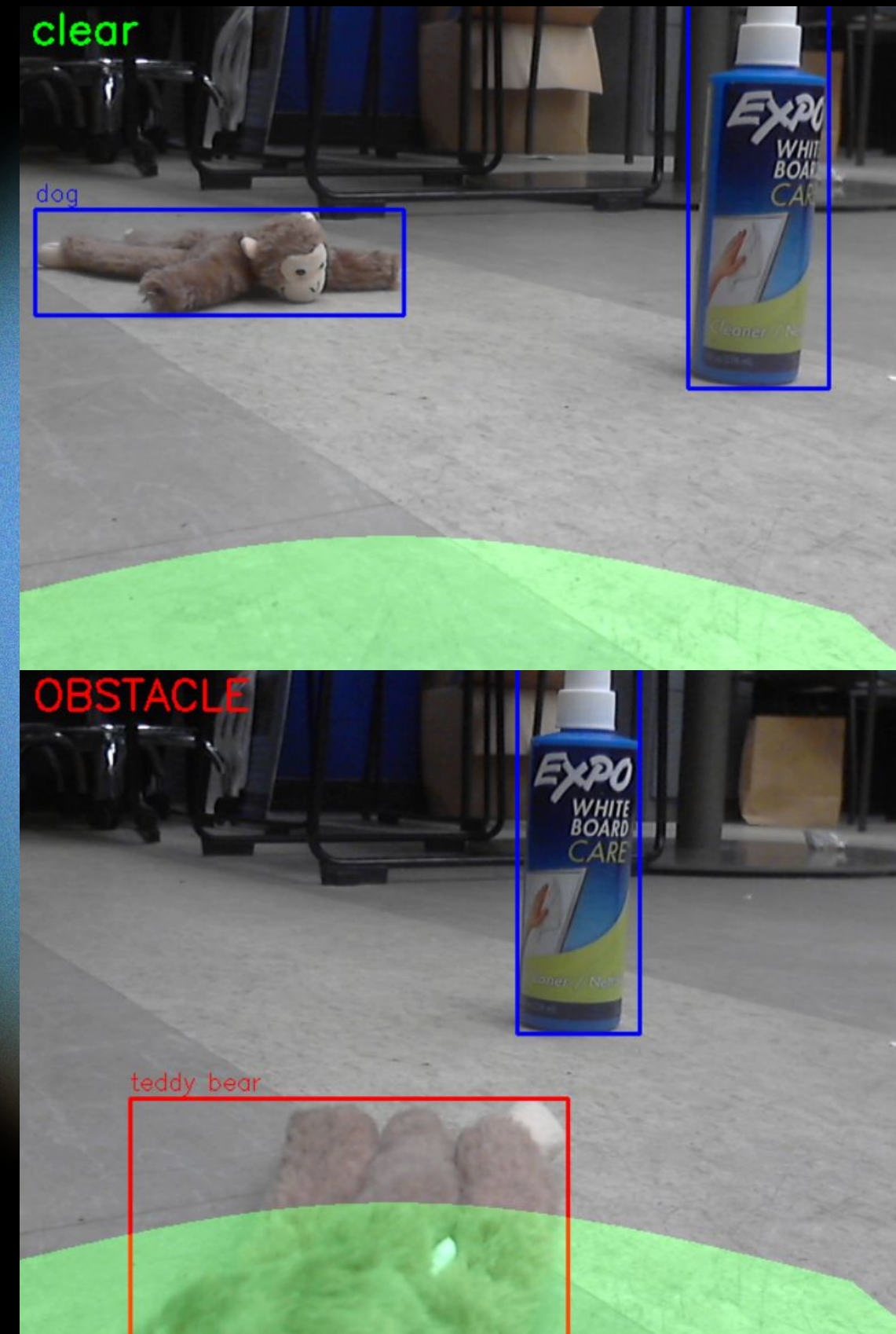
Scan taken of a room

Moving Parts



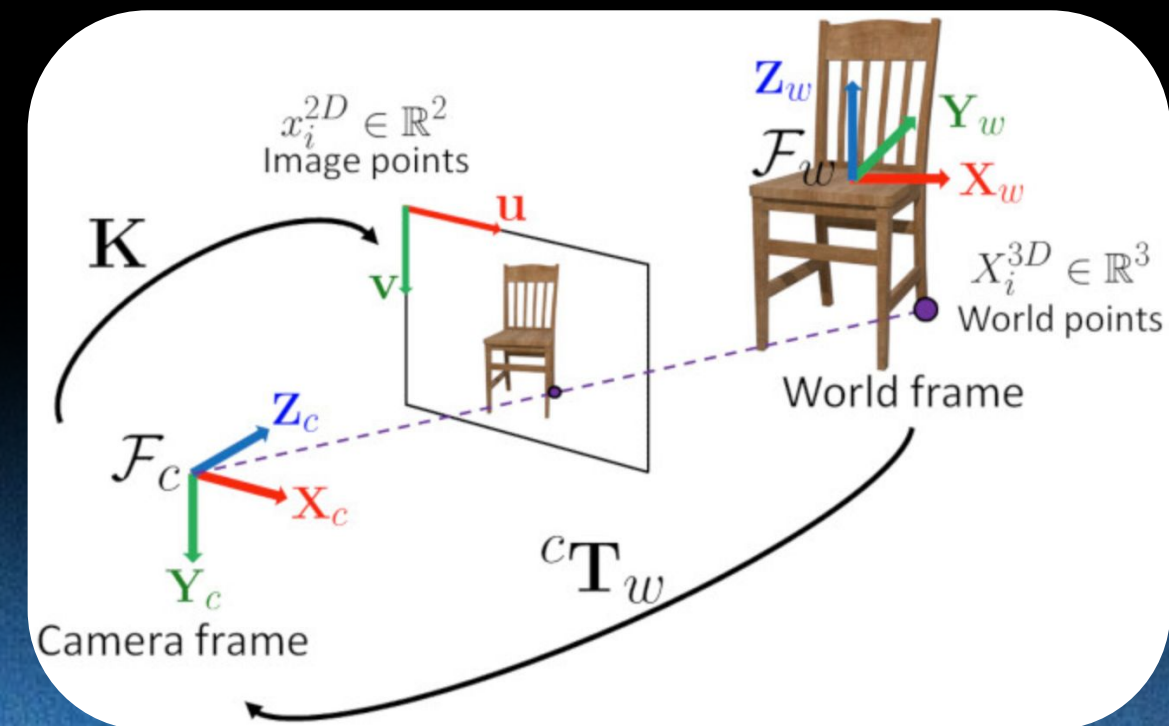
YOLO v8

- You Only Look Once
- A CNN made for various image tasks
- Acts as our obstacle detector
- Fast and efficient



Face Tracking

- Face landmarking accomplished through Google's Mediapipe model
- Model estimates facial landmarks
- Camera must be calibrated to account for lens distortion
- OpenCV used to solve the Perspective-n-Point problem



Conclusion

- We made a localized prototype of a wheelchair that removes the need for precise physical input from the user with a multimodal program comprised of
 - LiDAR
 - Camera
 - EEG
- We successfully built a prototype for our target population, giving them more autonomy and control. Addressing the issues of previous designs
- Next step: a full-scale, robust autonomous vehicle capable of supporting a human passenger

Demo and UI

L45deg

R45deg

L90deg

R90deg



Worker Status

Eyetrack

Motor

YOLO

Pathfind

PathCam

Classify

LIDAR

Camera Feed

Eyetracker

Pathfinding

Mirror

Gaze

Pos: (0.620, 0.169) | Face: ✓

Eyetracker

Eye: 1 | Path: 0

Smooth: 0.50

Motor

State: DRIVING

Target: +3.2° 5160mm

STOP

Pathfinding

→ +3.6° | 14928mm

Obs: LIDAR ok | YOLO ok

Classifier

Prediction: IDLE

Confidence: 100%

Demo Classifier

SAFETY KEYS ACTIVE [m for EEG]

Current: MOVE

MOVE

IDLE

JAW

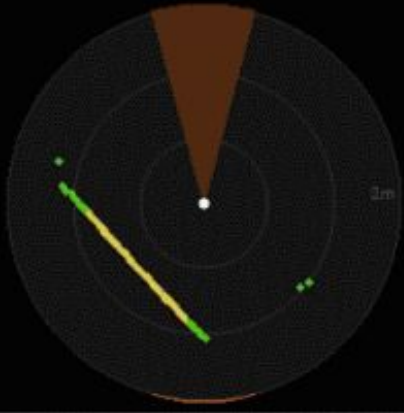
move

idle

jaw

dench

m toggle



Demo and UI



Turn and drive



YOLO Emergency Stop



Lidar Emergency Stop



Turn 45°

Thank you